Perform Exploratory Data Analysis (EDA) using Matplotlib to visualize trends, patterns, and anomalies in a given dataset. Generate bar charts and line graphs to uncover insights. Present findings through a visualization dashboard or report, summarizing key observations.

```
import pandas as pd

df_posts = pd.read_csv('/content/drive/MyDrive/SMA_DATASET/merged_questions.csv')
df_comments = pd.read_csv('/content/drive/MyDrive/SMA_DATASET/merged_comments.csv')
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
df_posts.head()
```

	Title	User	Relative time	Date	Time	Score	Link	Question
0	Ask HN: What are you working on? (February 2025)	david927	23 hours ago	2025-02-23	23:00:16	203	https://news.ycombinator.com/item?id=43154065	What are you working on? Any new ideas that y...
1	Ask HN: Former devs who can't get a job, what ...	throw81398475	7 hours ago	2025-02-24	20:22:37	20	https://news.ycombinator.com/item?id=43164498	That's all.Savings are getting low, and I'm go...
2	Ask HN: My father passed away. How do deal wit...	gogo61	4 hours ago	2025-02-24	23:37:25	9	https://news.ycombinator.com/item?id=43166256	It has been one month since he passed and i ha...
3	Tell HN: Five random IndieWeb blog links on	susam	1 day ago	2025-	10:55:47	44	https://news.ycombinator.com/item?	Hello HN! I believe some of you might

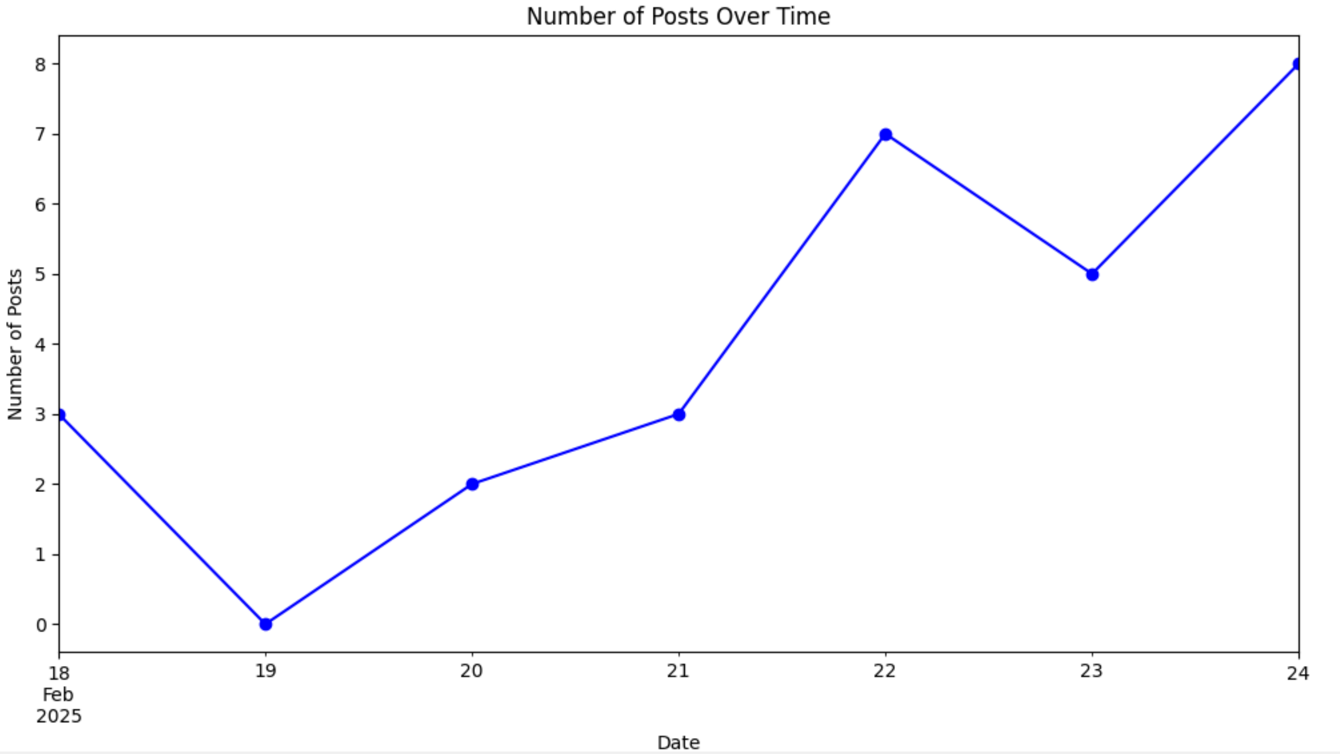
```
df_comments.head()
```

	Link	Author	Comment	Date	Time
0	https://news.ycombinator.com/item?id=43154065	jtwaleson	I'm creating an infinite canvas that has all y...	2025-02-24	08:27:29
1	https://news.ycombinator.com/item?id=43154065	abakker	Man...If you built this for large mainframe co...	2025-02-24	16:39:33
2	https://news.ycombinator.com/item?id=43154065	Cthulhu_	On a smaller scale it reminds me of the origin...	2025-02-24	13:21:21
3	https://news.ycombinator.com/item?id=43154065	jtwaleson	Cool. My long-term vision for software develop...	2025-02-24	18:04:44
4	https://news.ycombinator.com/item?id=43154065	Cerium	Sometimes I daydream that we moved beyond text...	2025-02-24	22:00:45

```
import matplotlib.pyplot as plt
import seaborn as sns
```

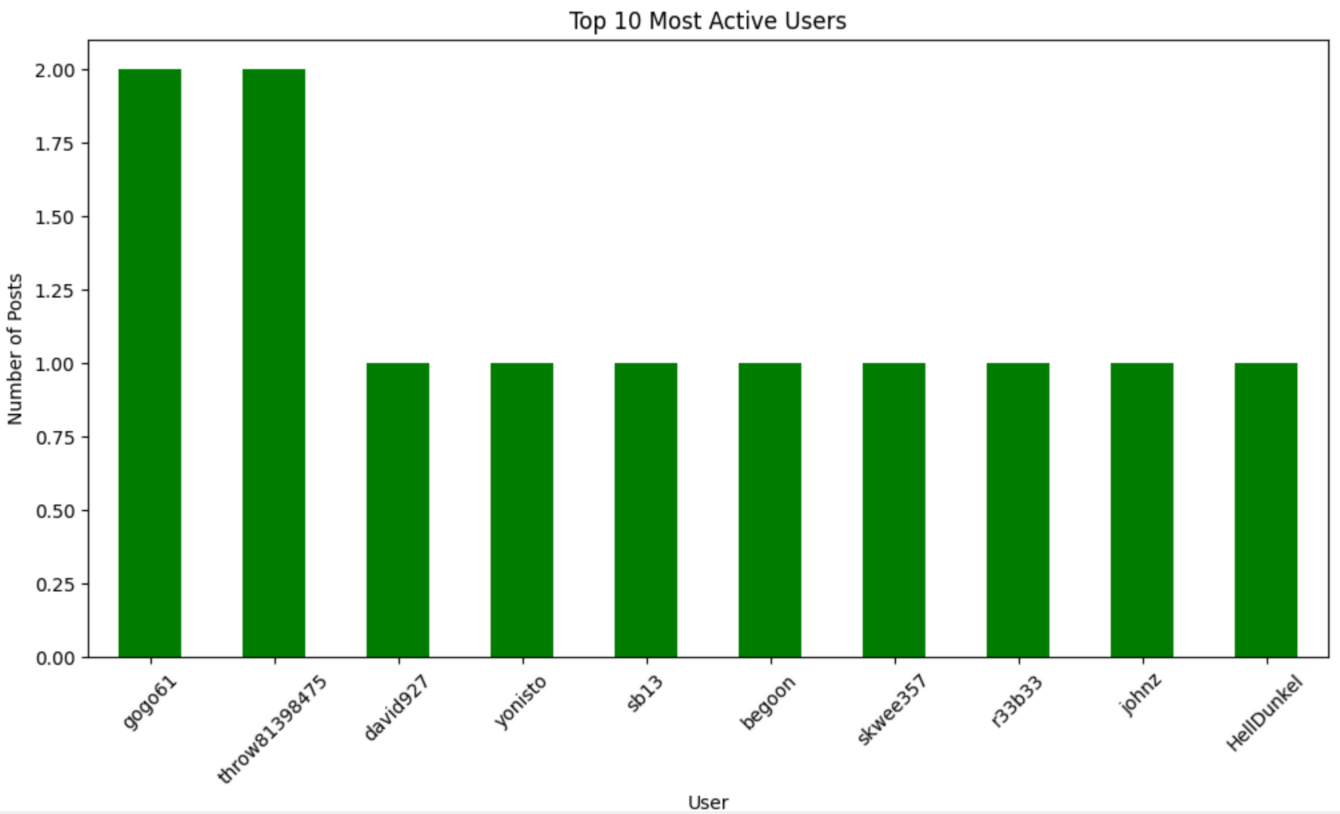
A. Posts Over Time (Line Chart)

```
df_posts['Date'] = pd.to_datetime(df_posts['Date'])
df_posts.set_index('Date', inplace=True)

plt.figure(figsize=(12,6))
df_posts.resample('D').size().plot(kind='line', color='blue', marker='o')
plt.title('Number of Posts Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Posts')
plt.show()
```



B. Most Active Users (Bar Chart)

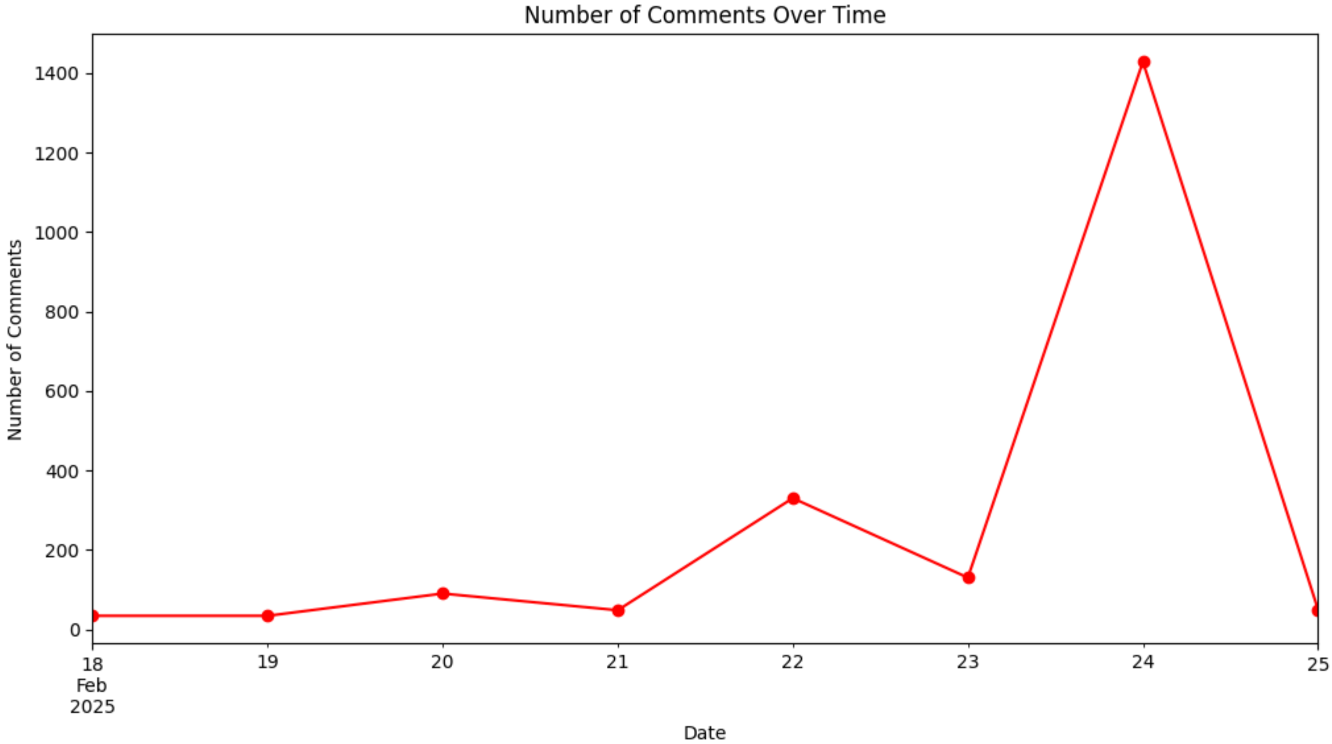
```
plt.figure(figsize=(12,6))
df_posts['User'].value_counts().head(10).plot(kind='bar', color='green')
plt.title('Top 10 Most Active Users')
plt.xlabel('User')
plt.ylabel('Number of Posts')
plt.xticks(rotation=45)
plt.show()
```



C. Comments Over Time (Line Chart)

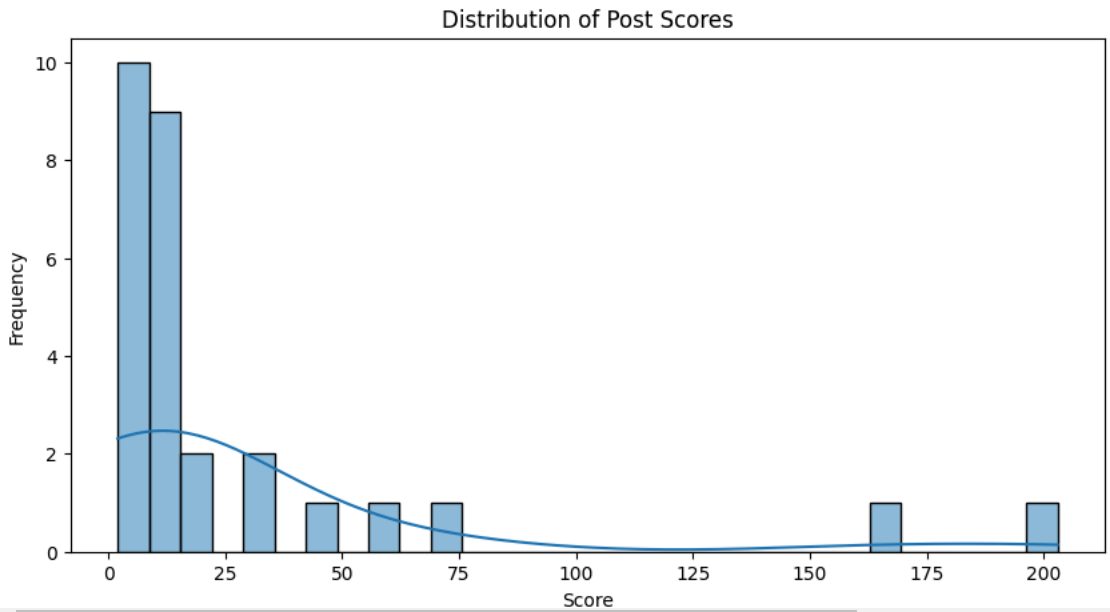
```
df_comments['Date'] = pd.to_datetime(df_comments['Date'])
df_comments.set_index('Date', inplace=True)
```

```
plt.figure(figsize=(12,6))
df_comments.resample('D').size().plot(kind='line', color='red', marker='o')
plt.title('Number of Comments Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Comments')
plt.show()
```



D. Distribution of Scores (Histogram)

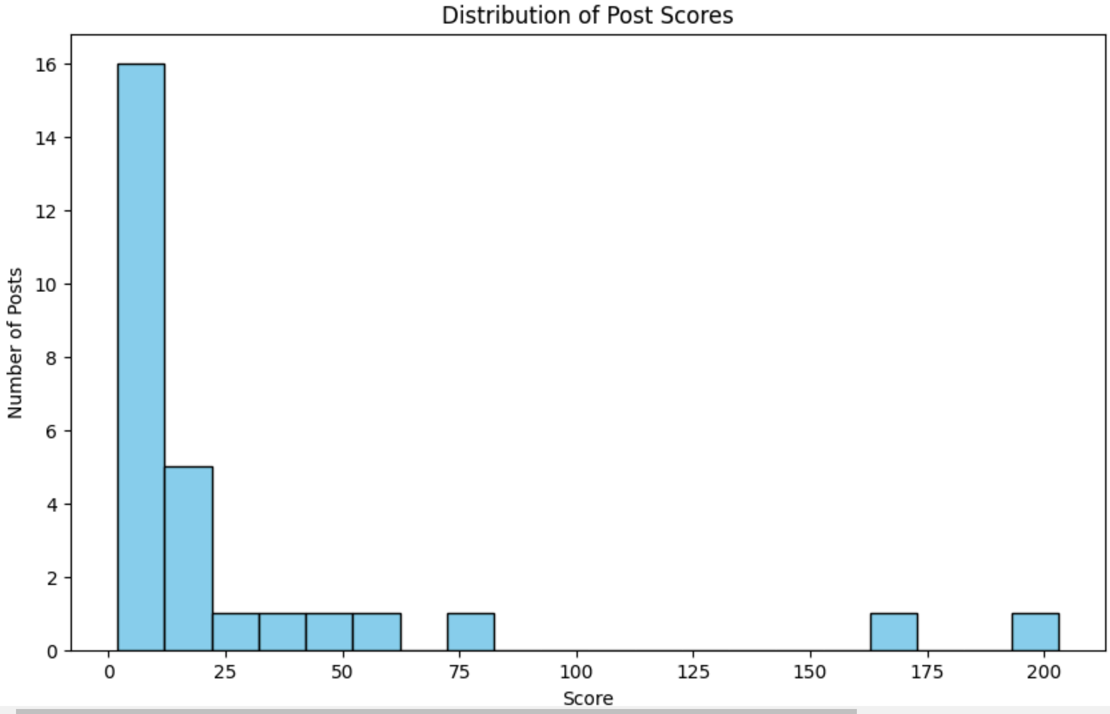
```
plt.figure(figsize=(10,5))
sns.histplot(df_posts['Score'], bins=30, kde=True)
plt.title('Distribution of Post Scores')
plt.xlabel('Score')
plt.ylabel('Frequency')
plt.show()
```



E. Distribution of Post Scores

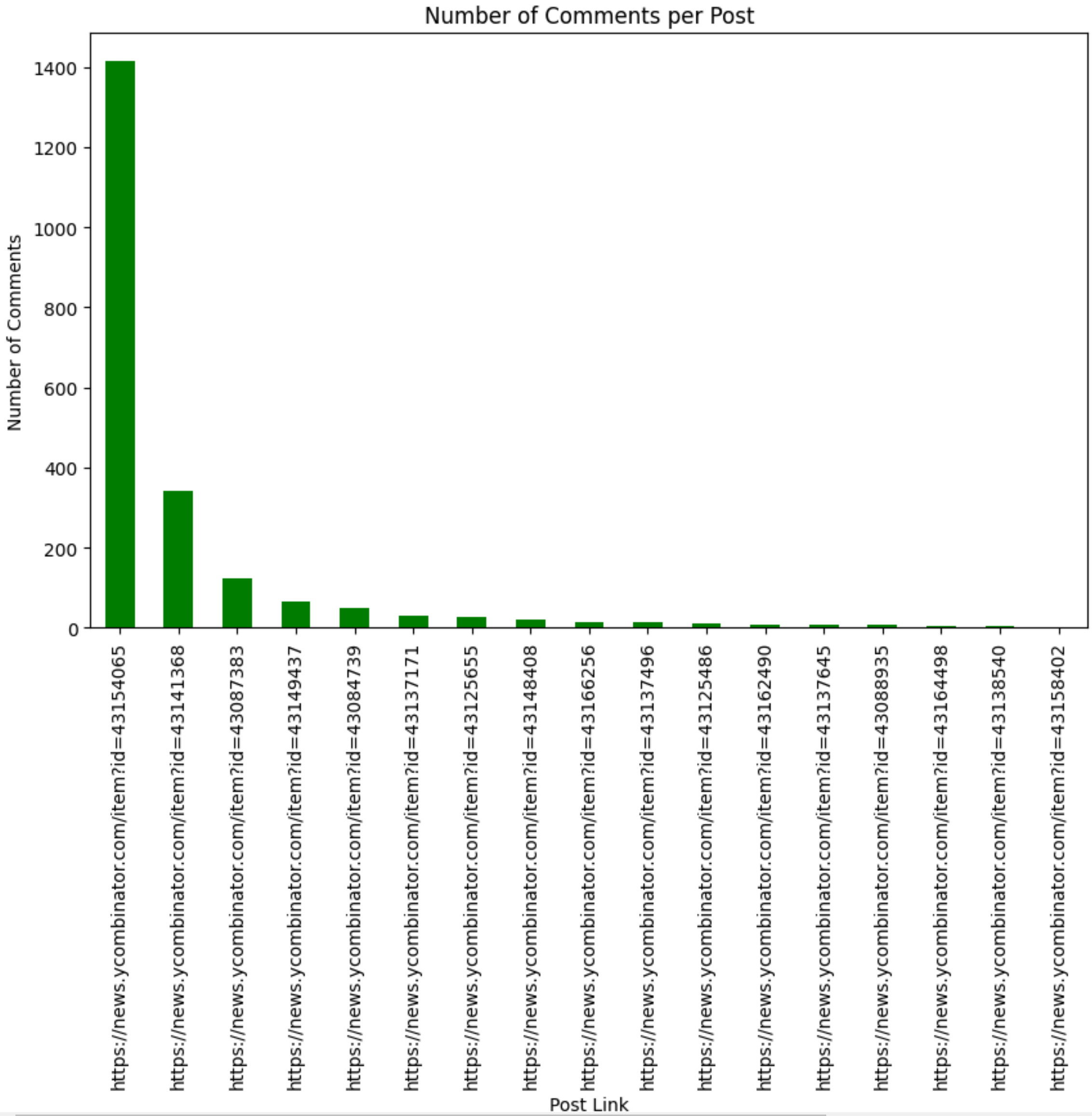
```
import pandas as pd
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(10, 6))
plt.hist(df_posts['Score'], bins=20, color='skyblue', edgecolor='black')
plt.title('Distribution of Post Scores')
plt.xlabel('Score')
plt.ylabel('Number of Posts')
plt.show()
```



F. Number of Comments per post

```
post_comment_counts = df_comments['Link'].value_counts()
plt.figure(figsize=(10, 6))
post_comment_counts.plot(kind='bar', color='green')
plt.title('Number of Comments per Post')
plt.xlabel('Post Link')
plt.ylabel('Number of Comments')
plt.show()
```



G: Scatter Plot: Post Scores vs. Number of Questions

```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Score', y='Question', data=df_posts, color='blue')
plt.title('Scatter Plot: Post Scores vs. Number of Questions')
plt.xlabel('Post Score')
plt.ylabel('Number of Questions')
plt.show()
```

```
comments_per_link = df_comments.groupby('Link')['Comment'].count().reset_index()
print("Number of comments for each link:")
print(comments_per_link)
```

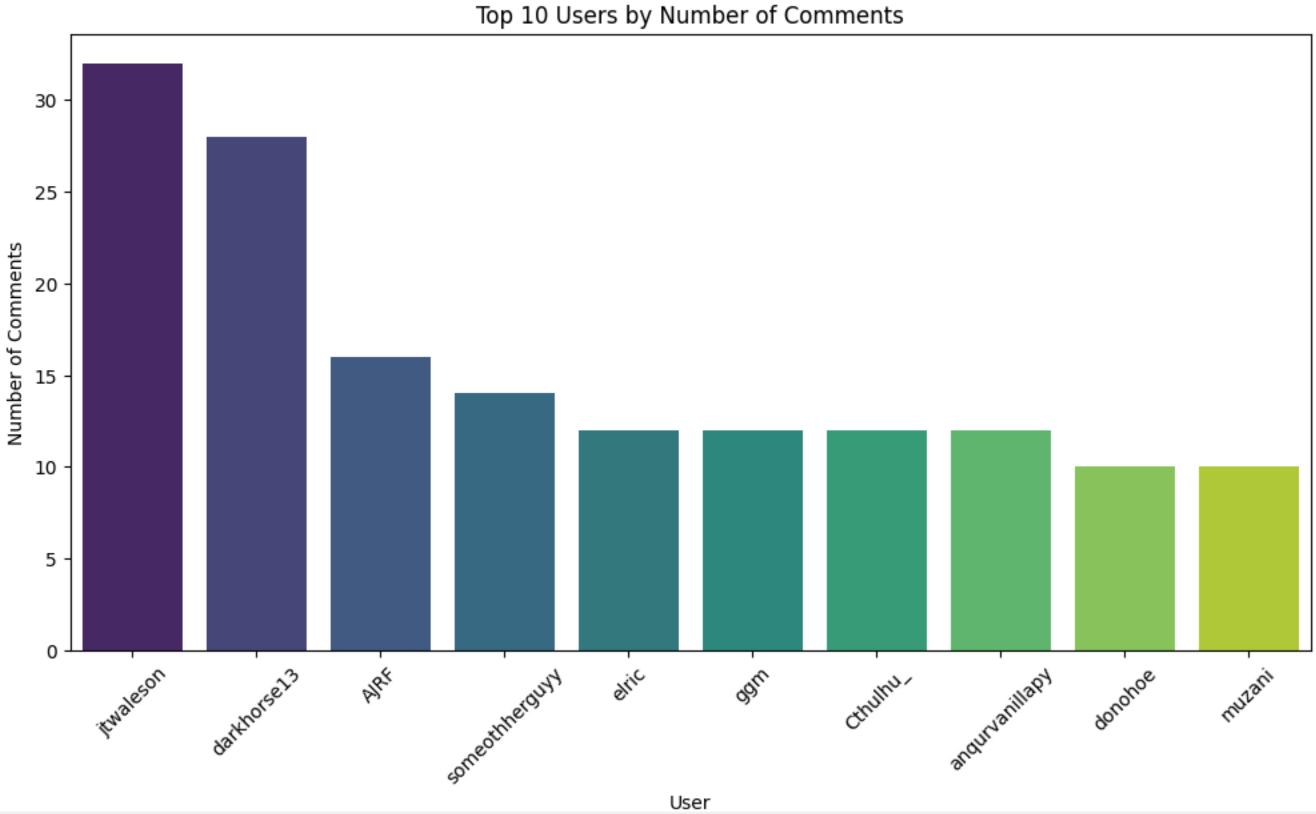
```
Number of comments for each link:
                                             Link  Comment
0   https://news.ycombinator.com/item?id=43084739       50
1   https://news.ycombinator.com/item?id=43087383      124
2   https://news.ycombinator.com/item?id=43088935        6
3   https://news.ycombinator.com/item?id=43125486       10
4   https://news.ycombinator.com/item?id=43125655       28
5   https://news.ycombinator.com/item?id=43137171       30
6   https://news.ycombinator.com/item?id=43137496       14
7   https://news.ycombinator.com/item?id=43137645        6
8   https://news.ycombinator.com/item?id=43138540        4
9   https://news.ycombinator.com/item?id=43141368      342
10  https://news.ycombinator.com/item?id=43148408       22
11  https://news.ycombinator.com/item?id=43149437       66
12  https://news.ycombinator.com/item?id=43154065     1414
13  https://news.ycombinator.com/item?id=43158402        2
14  https://news.ycombinator.com/item?id=43162490        8
15  https://news.ycombinator.com/item?id=43164498        4
```

16 https://news.ycombinator.com/item?id=43166256 14

H: Top 10 Users by Number of Comments

```
comments_per_link = df_comments.groupby('Link')['Comment'].count().reset_index()

merged_df = pd.merge(df_posts, comments_per_link, on='Link', how='left')

user_comment_counts = df_comments['Author'].value_counts().head(10)
plt.figure(figsize=(12, 6))
sns.barplot(x=user_comment_counts.index, y=user_comment_counts.values, palette='viridis')
plt.title('Top 10 Users by Number of Comments')
plt.xlabel('User')
plt.ylabel('Number of Comments')
plt.xticks(rotation=45)
plt.show()
```

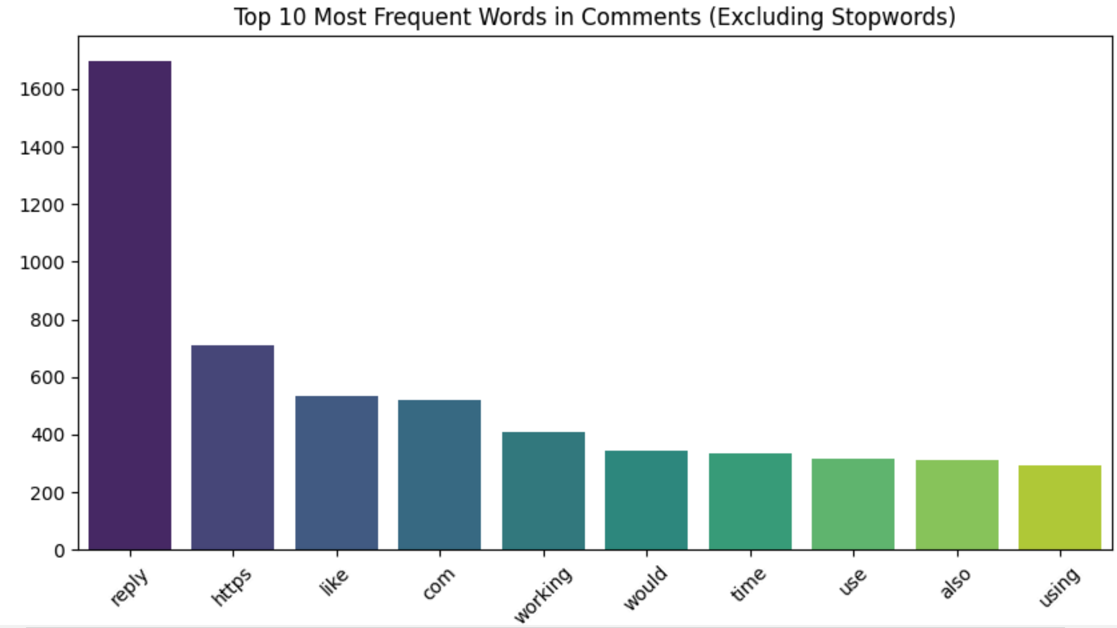
```
<ipython-input-15-1ab722906845>:7: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `le

  sns.barplot(x=user_comment_counts.index, y=user_comment_counts.values, palette='viridis')
```



I: Top 10 Most Frequent Words in Comments (Excluding Stopwords)

```
from collections import Counter
import re
import nltk
from nltk.corpus import stopwords

nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

all_words = [word for comment in df_comments['Comment'].dropna()
             for word in re.findall(r'\b[a-z]+\b', comment.lower())
             if word not in stop_words]

top_words = Counter(all_words).most_common(10)
words, counts = zip(*top_words)

plt.figure(figsize=(10,5))
sns.barplot(x=list(words), y=list(counts), palette='viridis')
plt.title('Top 10 Most Frequent Words in Comments (Excluding Stopwords)')
plt.xticks(rotation=45)
plt.show()
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
<ipython-input-25-65c8e3d6115a>:17: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `le

  sns.barplot(x=list(words), y=list(counts), palette='viridis')
```



J: Word Cloud for Comments

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt

comments = df_comments['Comment'].str.cat(sep=' ')

wordcloud = WordCloud(width=800, height=400, background_color='white').generate(comments)

plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Word Cloud for Comments')
plt.show()
```

K: Scatter Plot: Number of Comments vs. Length of Question

```
df_posts['Question_Length'] = df_posts['Title'].astype(str).apply(len)
comments_count = df_comments.groupby('Link').size().reset_index(name='Num_Comments')
merged = df_posts.merge(comments_count, on='Link', how='left').fillna(0)

sns.scatterplot(x=merged['Question_Length'], y=merged['Num_Comments'], alpha=0.7)
plt.title('Number of Comments vs. Length of Question')
plt.xlabel('Question Length')
```